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Guidance

Maths Functional Skills: subject content

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Applies to England

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Introduction

This guidance sets out the:

- purpose of Functional Skills in maths
- learning aims and outcomes
- subject content at:
  - Entry Levels 1, 2 and 3
  - Levels 1 and 2

A key aim for Functional Skills in maths is that they enable the student to gain:

- confidence
- fluency
- a positive attitude toward maths

The student will convey their confidence in using mathematics when they can:

- demonstrate a sound grasp of mathematical knowledge and skills
- apply it to solve mathematical problems

Awarding organisation specifications should encourage teachers to emphasise the interconnectedness of the 3 different areas of maths set out in this content:

- numbers and the number system
- common measures, shape and space
- information and data

At each level, there is an increase in the:

- difficulty of mathematical problem solving
- number and extent of connections made within the content

Mathematical problem-solving is an important aspect of Functional Skills, but it is also vital that the underpinning knowledge and skills[footnote 1] required (e.g. the use of times tables) can be demonstrated in their own right, both with and without a calculator. Awarding organisation specifications should encourage teachers to ensure that core knowledge and skills are secure in their students.

Throughout this guidance, mathematical problem solving is conveyed via the following terms:

- simple (Entry Level)
- straightforward (Level 1)
- complex (Level 2)

Each term relevant to that level is explained in the subject content sections. In interpreting the content, awarding organisations should note that the content at each level of qualification subsumes and builds upon the content at lower levels.

Purpose

Functional Skills qualifications should provide:

- reliable evidence of a student's achievements against demanding content that is relevant to the workplace
- assessment of their underpinning knowledge as well as their ability to apply this in different contexts
- a foundation for:
  - progression into employment
  - progression into further technical education

- developing skills for everyday life

In some contexts, Functional Skills qualifications will play a part in the government's accountability systems.

Functional Skills specifications should enable the student to develop behaviours such as persistence and logical thinking as they apply to mathematical tools and approaches.

Purpose: Entry Levels 1, 2 and 3

The purpose of Functional Skills in maths at Entry Levels 1, 2 and 3 is to demonstrate:

- a sound grasp of the underpinning skills and basics of mathematical skills appropriate to the level
- the ability to apply mathematical thinking to solve simple problems in familiar situations

Achievement of these qualifications can provide the skills for further study at Levels 1 and 2.

Purpose: Levels 1 and 2

The purpose of Functional Skills in maths at Levels 1 and 2 is to provide a qualification for work, study and life. Achievement of the qualifications demonstrates:

- a sound grasp of mathematical skills at the appropriate level
- the ability to apply mathematical thinking effectively to solve problems successfully in the workplace and other real-life situations

Learning aims and outcomes: Entry Levels 1 to 3

Functional Skills qualifications at these levels should:

- enable students to become confident in their use of fundamental mathematical knowledge and skills, as described through the content
- indicate that they can demonstrate their understanding by applying their knowledge and skills to solve simple mathematical problems or carry out simple tasks

Subject content: Entry Level 1

Using numbers and the number system – whole numbers
Students at Entry Level 1 are expected to be able to:

1. read, write, order and compare numbers up to 20
2. use whole numbers to count up to 20 items including zero
3. add numbers which total up to 20 and subtract numbers from numbers up to 20
4. recognise and interpret the symbols +, – and = appropriately

Using common measures, shape and space
Students at Entry Level 1 are expected to be able to:

5. recognise coins and notes, and write them in numbers with the correct symbols (£ and p), where these involve numbers up to 20

6. read 12-hour digital and analogue clocks in hours

7. know the number, name and sequence of:

- days in a week
- months
- the seasons

8. describe and make comparisons in words between measures of items, including:

- size
- length
- width

- height
- weight
- capacity

9. identify and recognise common 2-dimensional (2-D) and 3-dimensional (3-D) shapes, including a:

- circle
- cube
- rectangle (includes squares)
- triangle

10. use everyday positional vocabulary to describe position and direction, including:

- left
- right
- in front
- behind
- under
- above

Handling information and data
Students at Entry Level 1 are expected to able to:

11. read numerical information from lists

12. sort and classify objects using a single criterion

13. read and draw simple charts and diagrams, including a:

- tally chart
- block diagram
- graph

Solving mathematical problems and decision-making
Students at Entry Level 1 are expected to be able to use the knowledge and skills set out in the subject content section to:

- recognise a simple mathematical problem

- obtain a solution

A simple mathematical problem is one that requires working through one step or process.

At Entry Level 1, it is expected that students will be able to address individual problems, each of which draw on knowledge and skills from one mathematical content area:

- number and the number system
- common measures, shape and space
- information and data

They are expected to be able to:

- use given mathematical information
- recognise and use simple mathematical terms appropriate to Entry Level 1
- use the methods listed in the subject content section to produce, check and present results that make sense
- present appropriate explanations using:
  - numbers
  - measures
  - simple diagrams
  - simple charts
  - symbols
- provide a simple explanation for those results

The context for simple problems at this level should be familiar to all students and easily described.

Subject content: Entry Level 2

Using numbers and the number system – whole numbers, fractions and decimals

Students at Entry Level 2 are expected to be able to:

1. count reliably up to 100 items
2. read, write, order and compare numbers up to 200
3. recognise and sequence odd and even numbers up to 100
4. recognise and interpret the symbols +, – , x, ÷ and = appropriately
5. add and subtract 2-digit numbers
6. multiply whole numbers in the range 0 x 0 to 12 x 12 using times tables
7. know the number and sequence of:
    - hours in a day
    - weeks in a year
8. divide 2-digit whole numbers by single-digit whole numbers and express remainders
9. approximate by rounding to the nearest 10, and use this rounded answer to check results
10. recognise simple fractions (halves, quarters and tenths) of:

- whole numbers
- shapes

11. read, write and use decimals to one decimal place

Using common measures, shape and space

Students at Entry Level 2 are expected to be able to:

12. calculate money with pence up to one pound and in whole pounds of multiple items, and write the value using the correct symbols (£ or p)

13. read and record time in common date formats, understand hours from a 24-hour digital clock, and read the time displayed on an analogue clock in:

- hours
- half-hours
- quarter-hours

14. use metric measures of length, including:

- millimetres
- centimetres
- metres
- kilometres

15. use measures of weight, including:

- grams
- kilograms

16. use measures of capacity, including:

- millilitres
- litres

17. read and compare positive temperatures

18. read and use simple scales to the nearest labelled division

19. recognise and name 2-D and 3-D shapes, including:

- pentagons
- hexagons
- cylinders
- cuboids
- pyramids
- spheres

20. describe the properties of common 2-D and 3-D shapes, including:

- numbers of sides
- corners
- edges
- faces
- angles
- base

21. use appropriate positional vocabulary to describe position and direction, including:

- between
- inside
- outside
- middle
- below
- on top
- forwards
- backwards

Handling information and data

Students at Entry Level 2 are expected to be able to:

22. extract information from:

- lists
- tables
- diagrams
- bar charts

23. make numerical comparisons from bar charts

24. sort and classify objects using 2 criteria

25. take information from one format and represent the information in another format, including using a bar chart

Solving mathematical problems and decision-making

Students at Entry Level 2 are expected to be able to use the knowledge and/or skills set out in the subject content section to:

- recognise a simple mathematical problem
- obtain a solution

A simple mathematical problem is one that requires working through one step or process.

At Entry Level 2, it is expected that students will be able to address individual problems, each of which draw on knowledge and skills from one mathematical content area:

- number and the number system
- common measures, shape and space
- information and data

They are expected to be able to:

- use given mathematical information and methods, including:
  - numbers
  - symbols
  - simple diagrams
  - simple charts
- recognise, understand and use simple mathematical terms appropriate to Entry Level 2
- use the methods set out in the subject content section to produce, check and present results that make sense
- present explanations, using, as appropriate to Entry Level 2:
  - numbers
  - measures
  - simple diagrams
  - simple charts
  - symbols

The context for simple problems at this level should be familiar to all students and easily described.

Subject content: Entry Level 3

Using numbers and the number system – whole numbers, fractions and decimals

Students at Entry Level 3 are expected to be able to:

1. count, read, write, order and compare numbers up to 1,000
2. add and subtract using 3-digit whole numbers
3. divide 3-digit whole numbers by single- and double-digit whole numbers, and express remainders
4. multiply 2-digit whole numbers by single- and double-digit whole numbers
5. approximate by rounding numbers less than 1,000 to the nearest 10 or 100 and use this rounded answer to check results
6. recognise and continue linear sequences of numbers up to 100
7. read, write and understand thirds, quarters, fifths and tenths, including equivalent forms
8. read, write and use decimals up to 2 decimal places
9. recognise and continue sequences that involve decimals

Using common measures, shape and space

Students at Entry Level 3 are expected to be able to:

10. calculate with money using decimal notation, and express money correctly in writing in pounds and pence

11. round amounts of money to the nearest £1 or 10p

12. read, measure and record time using am and pm

13. read the time from analogue and 24-hour digital clocks in hours and minutes

14. use metric or imperial units to compare, to the nearest labelled or unlabelled division, measures of:

- length

- capacity
- weight
- temperature

15. compare metric measures of length, including:

- millimetres
- centimetres
- metres
- kilometres

16. compare measures of weight, including:

- grams
- kilograms

17. compare measures of capacity, including:

- millilitres
- litres

18. use a suitable instrument to measure:

- mass
- length

19. sort 2-D and 3-D shapes using properties including:

- lines of symmetry
- length
- right angles
- angles, including in rectangles and triangles

20. use appropriate positional vocabulary to describe position and direction, including:

- 8 compass points
- full-turns
- half-turns
- quarter-turns

Handling information and data

Students at Entry Level 3 are expected to be able to:

21. extract information and create frequency tables from:

- lists
- tables
- diagrams
- charts

22. interpret information to make comparisons and record changes from different formats, including:

- bar charts
- simple line graphs

23. organise and represent information in appropriate ways, including:

- tables
- diagrams
- simple line graphs
- simple bar charts

Solving mathematical problems and decision-making

Students at Entry Level 3 are expected to be able to use the knowledge and skills set out in the subject content section to:

- recognise a simple mathematical problem
- obtain a solution

A simple mathematical problem is one that requires working through one step or process.

At Entry Level 3, it is expected that students will be able to address individual problems, each of which draw on knowledge and/or skills from one mathematical content area:

- number and the number system
- common measures, shape and space
- information and data

They are expected to be able to:

- use given mathematical information, including:
  - numbers
  - symbols
  - simple diagrams
  - simple charts
- recognise, understand and use simple mathematical terms appropriate to Entry Level 3
- use the methods set out in the subject content section to produce, check and present accurate results that make sense to an appropriate level of accuracy
- present results with appropriate and reasoned explanation, using, as appropriate to Entry Level 3:
  - numbers
  - measures
  - simple diagrams
  - simple charts
  - symbols

The context for simple problems at this level should be familiar to all students.

Learning aims and outcomes at Levels 1 and 2

Functional Skills maths qualifications at these levels should:

- indicate that students can demonstrate, through appropriate reasoning and decision-making, their ability:
  - in mathematical skills
  - to apply these to solve realistic problems of increasing complexity
- introduce students to new areas of life and work so that they are exposed to concepts and problems that, while not of immediate concern, may be of value later
- enable students to develop an appreciation of the role played by maths at work and in life generally

Subject content: Level 1

Using numbers and the number system – whole numbers, fractions, decimals and percentages

Students at Level 1 are expected to be able to:

- count in steps of various sizes, including in negative numbers
- read, write and understand positive whole numbers to one million
- order and compare:
  - whole numbers of any size
  - fractions
  - ratios
  - decimals
- recognise the effect of multiplying and dividing by powers of:
  - 10
  - 100
  - 1,000
- identify, compare and extend a range of:
  - numerical patterns
  - spatial patterns

- use, understand and calculate with:
  - fractions
  - decimals
  - percentages
- calculate simple interest

They are specifically expected to be able to:

1. read, write, order and compare large numbers up to one million
2. recognise and use:
   - positive numbers
   - negative numbers
3. multiply and divide whole numbers and decimals by:
   - 10
   - 100
   - 1,000
4. use multiplication facts and make connections with division facts
5. use simple formulae expressed in words for:
   - one-step operations
   - 2-step operations
6. calculate the squares of:
   - one-digit numbers
   - 2-digit numbers
7. follow the order of precedence of operators
8. read, write, order and compare:
   - common fractions
   - mixed numbers
9. find fractions of whole-number:
   - quantities
   - measurements

10. read, write, order and compare decimals up to 3 decimal places
11. add, subtract, multiply and divide decimals up to 2 decimal places
12. approximate by rounding to:
    - a whole number
    - one decimal place
    - 2 decimal places
13. read, write, order and compare percentages in whole numbers

- calculate percentages of quantities, including:
  - simple percentage increases by 5%
  - simple percentage decreases by 5%
  - multiples thereof
- estimate answers to calculations using:
  - fractions
  - decimals
- recognise and calculate equivalences between:
  - common fractions
  - percentages
  - decimals
- work with:
  - simple ratio
  - direct proportions

Using common measures, shape and space

Students at Level 1 are expected to be able to:

- work out simple relationships between common units of measurement to define quantities
- use mathematical terms for position and direction
- apply and use calculations with common measures, including:
  - money

  - time
  - length
  - weight
  - capacity

- visualise, draw and describe 2-D and 3-D shapes
- use the properties of 2-D shapes in calculations

They are specifically expected to be able to:

14. calculate simple interest in multiples of 5% on amounts of money

15. calculate discounts in multiples of 5% on amounts of money

16. convert in the same system between units of:

- length
- weight
- capacity
- money
- time

17. recognise and make use of simple scales on:

- maps
- drawings

18. calculate the area and perimeter of simple shapes, including those that are made up of a combination of rectangles

19. calculate the volumes of:

- cubes
- cuboids

20. draw 2-D shapes and demonstrate an understanding of line symmetry and knowledge of the relative size of angles

21. interpret plans, elevations and nets of simple 3-D shapes

22. measure angles in degrees and use angles when describing:

- position
- direction

Handle information and data

Students at Level 1 are expected to be able to:

- select, construct and interpret a range of statistical diagrams in various contexts
- select and use methods and forms to present and describe outcomes
- extract and interpret information from:
  - tables
  - diagrams
  - charts
  - graphs
- apply simple statistics
- recognise features of charts to summarise and compare sets of data
- recognise and use the probability scale and interpret probabilities

They are specifically expected to be able to:

23. represent discrete data in tables, diagrams and charts, including:

- pie charts
- bar charts
- line graphs

24. group discrete data and represent grouped data graphically

25. find the mean and range of a set of quantities

26. understand probability on a scale from 0 (impossible) to 1 (certain) and use probabilities to compare the likelihood of events

27. use equally likely outcomes to find the probabilities of simple events and express them as fractions

Solving mathematical problems and decision-making

Students at Level 1 are expected to be able to use the knowledge and skills set out in the subject content section to:

- recognise a straightforward problem
- obtain a solution or solutions to a straightforward problem

A straightforward problem is one that requires students to work through either:

- one step or process
- more than one connected step or process

Individual problems are based on the knowledge and/or skills from the mathematical content areas:

- number and the number system
- common measures, shape and space
- information and data

At Level 1, it is expected that the student will be able to address individual problems, some of which draw upon a combination of any 2 of the mathematical content areas and require students to make connections between those content areas.

Students at Level 1 are expected to be able to:

- read, understand and use mathematical information and terms appropriate to Level 1
- address individual problems as described
- use knowledge and understanding to a required level of accuracy
- check the sense and reasonableness of answers
- analyse and interpret results in the context of the original problem
- present results with appropriate explanation and interpretation, demonstrating simple reasoning to support the process, and show consistency with the evidence presented

The context of individual problems at this level will require some comprehension in order for the student to be able to independently identify and

carry out an appropriate mathematical approach.

Subject content: Level 2

Using numbers and the number system – whole numbers, fractions, decimals, and percentages

Students at Level 2 are expected to be able to:

- use numbers of any size
- read, write and make use of:
  - positive integers of any size
  - negative integers of any size
- use, order and compare:
  - integers
  - fractions
  - decimals
  - percentages
  - ratios
- recognise the value of a digit in any:
  - whole number
  - decimal number
- use numerical and spatial patterns for a purpose and calculate with, and convert between, numbers written as fractions, decimals, percentages and ratios

They are specifically expected to be able to:

1. read, write, order and compare:
   - positive numbers of any size
   - negative numbers of any size

2. carry out calculations with numbers up to one million, using strategies to check answers, including:
    - estimation
    - approximation
3. evaluate expressions and make substitutions in given formulae in:
    - words
    - symbols
4. identify and know the equivalence between:
    - fraction
    - decimals
    - percentages
5. work out percentages of amounts and express one amount as a percentage of another
6. calculate:
    - percentage change (any size of increase and decrease)
    - original value after percentage change
7. order, add, subtract and compare amounts or quantities using:
    - proper fractions
    - improper fractions
    - mixed numbers
8. express one number as a fraction of another
9. order, approximate and compare decimals
10. add, subtract, multiply and divide decimals to 3 decimal places
11. understand and calculate using:
    - ratios
    - direct proportion
    - inverse proportion
12. follow the order of precedence of operators, including indices

Using measures, shape and space

Students at Level 2 are expected to be able to:

- handle relationships between measurements of various kinds
- use angles and co-ordinates when describing position and direction
- make use of geometric properties in calculations with 2-D and 3-D shapes, and understand the relationships between them

They are specifically expected to be able to:

13. calculate amounts of:

- money
- compound interest
- percentage increases
- percentage decreases
- discounts, including tax and simple budgeting

14. convert, using a conversion factor and conversion graph, between metric and imperial units of:

- length
- weight
- capacity

15. calculate using compound measures, including:

- speed
- density
- rates of pay

16. calculate the perimeter and area of 2-D shapes, including:

- triangles
- circles

- composite shapes, including non-rectangular shapes (formulae given, except for triangles and circles)

17. use formulae to find volumes and surface areas of 3-D shapes, including cylinders (formulae to be given for 3-D shapes other than cylinders)

18. calculate actual dimensions from scale drawings and create a scale diagram, given actual measurements

19. use co-ordinates in 2-D, positive and negative, to specify the positions of points

20. understand and use common 2-D representations of 3-D objects

21. draw 3-D shapes, including plans and elevations

22. calculate values of angles and/or co-ordinates with 2-D and 3-D shapes

Handling information and data

Students at Level 2 are expected to be able to:

- construct, interpret and evaluate a range of statistical diagrams
- calculate and interpret probabilities
- calculate, analyse, compare and interpret appropriate:
  - data sets
  - tables
  - diagrams
  - statistical measures such as common averages (mean, median, mode) and spread (range)
- use statistics to compare 2 sets of data
- identify patterns and trends from data
- recognise simple correlation

They are specifically expected to:

23. calculate the median and mode of a set of quantities

24. estimate the mean of a grouped frequency distribution from discrete data

25. use the mean, median, mode and range to compare 2 sets of data

26. work out the probability of combined events, using diagrams and tables, including 2-way tables

27. express probabilities as:

- fractions
- decimals
- percentages

28. draw and interpret scatter diagrams, and recognise:

- positive correlation
- negative correlation

Solving mathematical problems and decision-making

Students at Level 2 are expected to be able to use the knowledge and skills set out in the subject content section to:

- recognise a complex mathematical problem
- obtain a solution or solutions

A complex mathematical problem is one that requires:

- a multi-step process
- planning and working through at least 2 connected steps or processes

Individual problems are based on a combination of the knowledge and skills from the mathematical content areas:

- number and the number system
- common measures, shape and space
- information and data

At Level 2, it is expected that the student will be able to address individual problems, some of which draw on a combination of all 3 content areas and require students to make connections between them.

Students at Level 2 are expected to be able to:

- read, understand and use mathematical information and terms
- address individual problems as described
- use knowledge and understanding to a required level of accuracy
- identify suitable operations and calculations to generate results
- analyse and interpret results in the context of the original problem
- check the sense and reasonableness of answers
- present and explain results clearly and accurately, demonstrating reasoning to support the process, and show consistency with the evidence presented

The context of individual problems at this level will require interpretation and analysis in order for the student to be able to independently identify and carry out an appropriate mathematical process or processes.

Explanation behind the use of the term mathematical problem solving (for information)

Mathematical problem-solving is a core element of Functional Skills mathematics, though underpinning knowledge will also be tested in its own right.

Problem-solving should not seek to obscure or add additional complexity beyond the level of the qualification.

Defining what problem-solving means in the context of examinations is challenging. In discussing this issue, a working group for higher-level qualifications[footnote 2] suggested that considering attributes of problem-solving was a way forward. They came to a consensus regarding a range of attributes typical of problem-solving questions. They emphasised that not all – in fact, often just one – of these attributes may be necessary to be present within a single task in order to consider it as involving problem-solving[footnote 3]. This is especially pertinent when considering the difference in the intended level of challenge between students studying for higher-level qualifications compared to Functional Skills.

The attributes, of which one or more may be present in a single task to consider it as problem-solving, are tasks:

- that have little or no scaffolding – the student is given little guidance beyond a start and finish point, and the questions do not explicitly state the mathematical processes required for the solution
- that provide for multiple representations, such as the use of a sketch or a diagram, as well as calculations
- where the information is not given in mathematical form or in mathematical language, or there is a need for the results to be interpreted or the methods evaluated – for example, in a real-world context
- have a variety of techniques that could be used
- where the solution requires an understanding of the processes involved, rather than just application of the techniques
- require 2 or more mathematical processes or may require different parts of mathematics to be brought together to reach a solution[footnote 4]

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1. The ability to do maths when not as part of a problem.
2. See A level mathematics working group report (2015) (https://assets.publishing.service.gov.uk/media/5a7f203540f0b6230268d970/a-level-mathematics-working-group-report.pdf), Ofqual/15/5789, pages 4-5
3. Problem-solving tasks are tasks that focus primarily on the assessment of problem-solving or a set of requirements focusing on one problem. They may be broken down into steps or parts (that is, items). See the report Problem-Solving in Mathematics (https://royalsociety.org/news-resources/publications/2016/problem-solving-through-mathematics-realising-the-vision-through-better-assessment/), The Royal Society's Advisory Committee on Mathematics Education (ACME), June 2016.
4. Not all of these attributes would be required within a single task to establish it as problem-solving. Neither does the presence of one or more attributes within a task automatically imply problem-solving is taking place.

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